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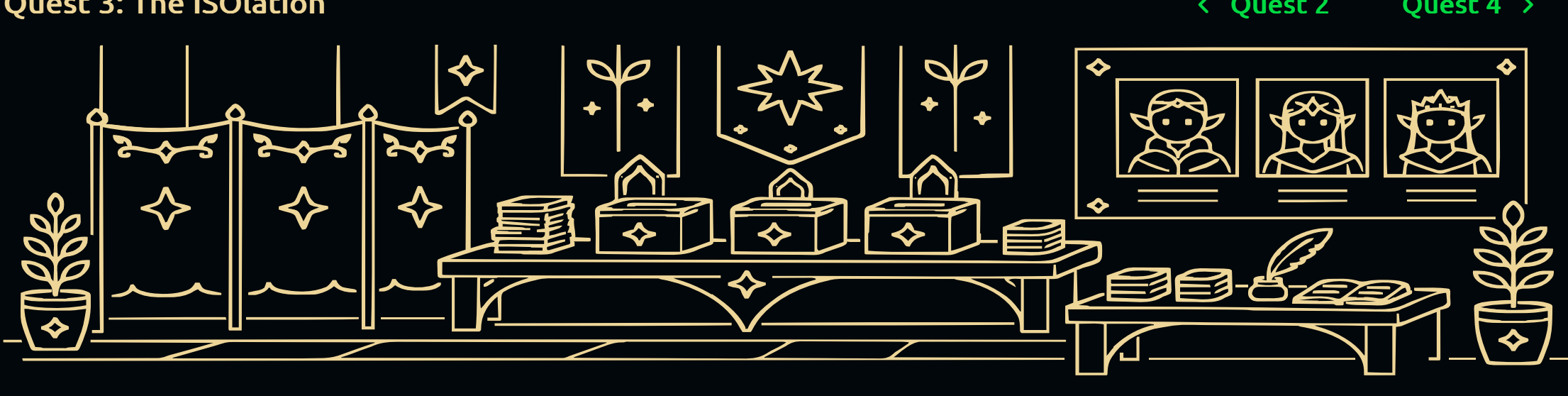
Quest 3: The ISOlation

‹ Quest 2 Quest 4 ›



Part I

Story section

With his research finally complete, Gimlock happily joined our expedition. The smile on his face lasted exactly until he learned that our next stop would be the Elven Capital. We still needed the elves' help if we were to continue our journey to Nulldor.

Gimlock stopped walking immediately. Like every self-respecting dwarf, he had never trusted elves. He muttered something about elves forming a committee before they could hammer in a nail. After a little more grumbling, he reluctantly agreed to continue.

When we finally arrived, we discovered that the elves were in the middle of electing three members of their Grand Council. Most of the votes had already been counted, but the counting itself had become rather complicated. There was nobody in charge for us to speak to yet, so we decided we might as well help finish the count.

Elves had a remarkable talent for over-engineering even the simplest ideas, and their number system was no exception. Instead of writing numbers in the ordinary decimal notation, they had introduced a collection of different standard representations known as `ISO` standards.

Unfortunately, over the centuries they had invented so many different ISO standards that nobody could agree which one was actually the standard. The obvious solution, of course, was to declare all of them equally standard. By then, nobody could even remember what the letters "ISO" originally stood for.

Every vote was written as two values separated by a space: `ISO-3210 7654`. The part after `ISO-` defined the standard, while the second value was the vote itself written using that standard. **Each digit of the standard specified the weight base** used for the corresponding digit of the vote, starting from the least significant digit. Just like in ordinary positional number systems, the rightmost weight base was raised to the power of 0, the next one to the power of 1, then 2, and so on.

Every vote contained a number representing how much the voter supported the chosen candidate, written in one of the many elven ISO standards. We received the votes cast for one of the candidates (your notes). Our task was to **convert every vote back into good old decimal notation, sum all the values**, and report the total score.

Example based on the following notes:

```
ISO-3210 7654
ISO-8742 2468
ISO-1111 1111
```

The individual votes from the list could be converted into decimal notation as follows:

```
ISO-3210 7654 = 222

ISO-3210
    ||||
    |||+-----------------+
    ||+-----------+      |
    |+-----+      |      |
    |      |      |      |
  7*3³ + 6*2² + 5*1¹ + 4*0⁰ = 189 + 24 + 5 + 4 = 222
  |      |      |      |
  |+-----+      |      |
  ||+-----------+      |
  |||+-----------------+
  ||||
  ||||
  7654
```

```
ISO-8742 2468 = 1252

ISO-8742
    ||||
    |||+-----------------+
    ||+-----------+      |
    |+-----+      |      |
    |      |      |      |
  2*8³ + 4*7² + 6*4¹ + 8*2⁰ = 1024 + 196 + 24 + 8 = 1252
  |      |      |      |
  |+-----+      |      |
  ||+-----------+      |
  |||+-----------------+
  ||||
  ||||
  2468
```

```
ISO-1111 1111 = 4

ISO-1111
    ||||
    |||+-----------------+
    ||+-----------+      |
    |+-----+      |      |
    |      |      |      |
  1*1³ + 1*1² + 1*1¹ + 1*1⁰ = 1 + 1 + 1 + 1 = 4
  |      |      |      |
  |+-----+      |      |
  ||+-----------+      |
  |||+-----------------+
  ||||
  ||||
  1111
```

The candidate's final score in the decimal number system was therefore: 222 + 1252 + 4 = **1478**

What was the candidate's total score in the decimal number system?

Your notes for this part: Copy Open Download

Part I solved with answer: **12065**

Check your progress

Part II

We finished counting the votes surprisingly quickly, so the elven officials immediately handed us the votes cast for another candidate (your notes).

The work followed the same rules as before, with just one complication. Quite often, **the number of digits in the ISO standard no longer matched the number of digits in the vote written using that standard.**

One of the officials explained that **an ISO standard represented a repeating pattern**. Whenever more positional weights were needed, the pattern repeated until the entire number was covered.

For example:

- `ISO-123 756423` could be treated in exactly the same way as `ISO-123123 756423`.
- `ISO-123 7564231` could be treated in exactly the same way as `ISO-3123123 7564231`.
  Note that **the pattern is repeated from right to left** so that it covers the entire number.
- `ISO-2 12345678` could be treated in exactly the same way as `ISO-22222222 12345678`.

Apart from that, nothing else had changed. We still had to convert every vote into decimal notation, add all the values together, and report the candidate's final score.

Example based on the following notes:

The individual votes from the list could be converted into decimal notation as follows:

```
ISO-3210 7654
ISO-8742 2468
ISO-1 1111
ISO-13 777
ISO-34 140856
ISO-789 2465112
ISO-2341 945
```

```
ISO-3210 7654 = 222

ISO-3210
    ||||
    |||+-----------------+
    ||+-----------+      |
    |+-----+      |      |
    |      |      |      |
  7*3³ + 6*2² + 5*1¹ + 4*0⁰ = 189 + 24 + 5 + 4 = 222
  |      |      |      |
  |+-----+      |      |
  ||+-----------+      |
  |||+-----------------+
  ||||
  ||||
  7654
```

```
ISO-8742 2468 = 1252

ISO-8742
    ||||
    |||+-----------------+
    ||+-----------+      |
    |+-----+      |      |
    |      |      |      |
  2*8³ + 4*7² + 6*4¹ + 8*2⁰ = 1024 + 196 + 24 + 8 = 1252
  |      |      |      |
  |+-----+      |      |
  ||+-----------+      |
  |||+-----------------+
  ||||
  ||||
  2468
```

```
ISO-1 1111 = 4

ISO-1111
    ||||
    |||+-----------------+
    ||+-----------+      |
    |+-----+      |      |
    |      |      |      |
  1*1³ + 1*1² + 1*1¹ + 1*1⁰ = 1 + 1 + 1 + 1 = 4
  |      |      |      |
  |+-----+      |      |
  ||+-----------+      |
  |||+-----------------+
  ||||
  ||||
  1111
```

```
ISO-13 777 = 77

ISO-313
    |||
    ||+-----------+
    |+-----+      |
    |      |      |
  7*3² + 7*1¹ + 7*3⁰ = 63 + 7 + 7 = 77
  |      |      |
  |+-----+      |
  ||+-----------+
  |||
  |||
  777
```

```
ISO-34 140856 = 1416

ISO-343434
    ||||||
    |||||+-----------------------------+
    ||||+-----------------------+      |
    |||+-----------------+      |      |
    ||+-----------+      |      |      |
    |+-----+      |      |      |      |
    |      |      |      |      |      |
  1*3⁵ + 4*4⁴ + 0*3³ + 8*4² + 5*3¹ + 6*4⁰ = 1416
  |      |      |      |      |      |
  |+-----+      |      |      |      |
  ||+-----------+      |      |      |
  |||+-----------------+      |      |
  ||||+-----------------------+      |
  |||||+-----------------------------+
  ||||||
  ||||||
  140856
```

```
ISO-789 2465112 = 1158390

ISO-9789789
    |||||||
    ||||||+-----------------------------------+
    |||||+-----------------------------+      |
    ||||+-----------------------+      |      |
    |||+-----------------+      |      |      |
    ||+-----------+      |      |      |      |
    |+-----+      |      |      |      |      |
    |      |      |      |      |      |      |
  2*9⁶ + 4*7⁵ + 6*8⁴ + 5*9³ + 1*7² + 1*8¹ + 2*9⁰ = 1158390
  |      |      |      |      |      |      |
  |+-----+      |      |      |      |      |
  ||+-----------+      |      |      |      |
  |||+-----------------+      |      |      |
  ||||+-----------------------+      |      |
  |||||+-----------------------------+      |
  ||||||+-----------------------------------+
  |||||||
  |||||||
  2465112
```

```
ISO-2341 945 = 102

ISO-341
    |||
    ||+-----------+
    |+-----+      |
    |      |      |
  9*3² + 4*4¹ + 5*1⁰ = 81 + 16 + 5 = 102
  |      |      |
  |+-----+      |
  ||+-----------+
  |||
  |||
  945
```

The candidate's final score in the decimal number system was therefore:
222 + 1252 + 4 + 77 + 1416 + 1158390 + 102 = **1161463**

What was the candidate's total score in the decimal number system?

Your notes for this part: Copy Open Download

Part II solved with answer: **20233862655**

Check your progress

Part III

After finishing the second count, we were handed one of the last sets of votes still waiting to be processed (your notes). It belonged to the candidate who had introduced **yet another format: ISO2**, so every vote in the set had been written using that standard. Gimlock stared at the papers for a moment before quietly announcing that he had discovered an entirely new reason to dislike elves.

In an ordinary ISO standard, several different numbers could represent exactly the same decimal value:

```
ISO-5322 1028 = 137
ISO-5322 1036 = 137
ISO-5322 1044 = 137
ISO-5322 1052 = 137
ISO-5322 1060 = 137
ISO-5322 1103 = 137
```

This had caused too much confusion, so ISO2 had been introduced to remove the ambiguity.

An ISO2 number was considered valid only if no smaller number represented the same value when interpreted using the same ISO2 standard. In other words, **it had to be the smallest possible numerical representation** of its value in that standard. Among the examples above, only `ISO2-5322 1028` was a valid ISO2-5322 representation of the value `137`.

We converted the votes into decimal notation as before, but this time we added only the values of valid ISO2 votes. **Every invalid vote was discarded before calculating the candidate's final score.**

Fortunately, the numbers in this final set were reasonably small, so checking them did not take as long as Gimlock had feared.

Example based on the following notes:

```
ISO2-3 23
ISO2-61 149
ISO2-978 99
ISO2-5887 5923
ISO2-76988 123749
ISO2-435657 929977
ISO2-3949 870737
```

The first vote represented the decimal value `9`. However, the smaller number `ISO2-3 9` represented exactly the same value, so `ISO2-3 23` **was not a valid ISO2 representation** and had to be discarded.

The second vote represented the decimal value `34`. The smaller number `ISO2-61 54` represented the same value, so `ISO2-61 149` **was also invalid.**

The following four votes were already the smallest possible representations of their values:

```
ISO2-978 99 = 72
ISO2-5887 5923 = 1220
ISO2-76988 123749 = 38826
ISO2-435657 929977 = 10869
```

They were therefore valid and their values were included in the final score.

The final vote represented the decimal value `54705`. However, the smaller number `ISO2-3949 280207` represented exactly the same value, so `ISO2-3949 870737` **was invalid** and had to be discarded.

The candidate's final score in decimal notation was therefore: 72 + 1220 + 38826 + 10869 = **50987**

What was the candidate's total score in the decimal number system after all invalid ISO2 votes had been discarded?

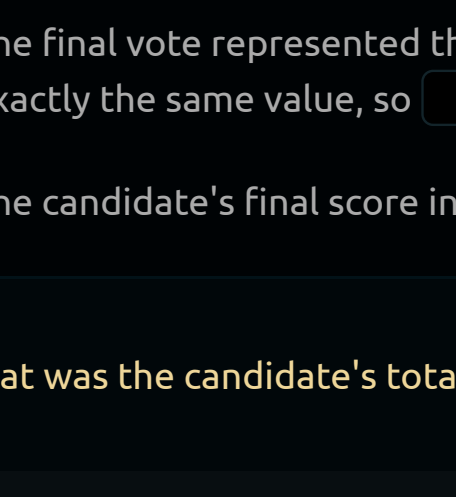
Your notes for this part: Copy Open Download

Part III solved with answer: **434479**

Check your progress

Puzzle solved! Don't stop now!

Post your solution, compare ideas, and help others grow on Reddit or join us for a chat on Discord



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